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RESEARCH ARTICLE

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Global Kilometer-Scale Climate Storylines Using Spectral Nudging

Amal John[1], Sebastian Beyer[1], Marylou Athanase[1], Antonio Sánchez-Benítez[1], Helge F. Goessling[1], Akil Hossain[1], Razvan Aguridan[2,3,4], Miguel Andrés-Martínez[1], Aina Gaya-Àvila[2], Suvarchal Kumar Cheedela[1], Philipp Geier[3], Rohit Ghosh[1], Ioan Hadade[3,4], Nikolay Koldunov[1], Sebastian Milinski[3,4], Matteo Nurisso[5], Xabier Pedruzo-Bagazgoitia[3,4], Thomas Rackow[1,3,4], Irina Sandu[3,4], Dmitry Sidorenko[1], Jan Streffing[1], Emanuele Vitali[6], and Thomas Jung[1,7]

[1]Alfred-Wegener-Institut, Helmholtz-Zentrum für Polar- und Meeresforschung (AWI), Bremerhaven, Germany, [2]Barcelona Supercomputing Center-Centro Nacional de Supercomputación (BSC-CNS), Barcelona, Spain, [3]European Centre for Medium-Range Weather Forecasts (ECMWF), Bonn, Germany, [4]European Centre for Medium-Range Weather Forecasts (ECMWF), Reading, UK, [5]Consiglio Nazionale delle Ricerche, Istituto di Scienze dell'Atmosfera e del Clima (CNR-ISAC), Turin, Italy, [6]CSC—IT Center for Science, Espoo, Finland, [7]Institute of Environmental Physics, University of Bremen, Bremen, Germany

Key Points:

- Combines the storylines approach using spectral nudging with km-scale modeling
- km-scale storylines deliver localized and granular information on extreme events

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

A. John,
amaljohn.met@gmail.com;
amal.john@awi.de

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Author Contributions:

Conceptualization: Amal John, Sebastian Beyer, Marylou Athanase, Antonio Sánchez-Benítez, Helge F. Goessling, Thomas Jung
Data curation: Amal John, Sebastian Beyer
Formal analysis: Amal John, Sebastian Beyer, Marylou Athanase, Antonio Sánchez-Benítez, Helge F. Goessling, Thomas Jung
Funding acquisition: Helge F. Goessling, Thomas Jung
Investigation: Amal John, Sebastian Beyer, Marylou Athanase,

Abstract

Effective climate adaptation benefits from detailed, actionable information on how specific extreme events are influenced by climate change. Here we present the first global, kilometer-scale coupled modeling framework that enables counterfactual reconstructions of recent weather events, including extremes, in different climates. The system is built on the IFS–FESOM model and applies scale-selective spectral nudging of the large-scale atmospheric circulation to ERA5 while allowing atmospheric small-scale dynamics and thermodynamic processes as well as the land surface, ocean, and sea ice to evolve freely across contrasting climate backgrounds. This enables the replay of recent events across different climate states, including the 1950s, present-day conditions, and a +2 K warmer climate. We demonstrate the system's skill in reproducing the observed variability on daily and longer timescales during the period 2017–2024, particularly in the extratropics. Furthermore, we evaluate its performance using high-quality observations, including data from the MOSAiC expedition. We further present climate change storylines for a European heatwave in July 2019 and a major European flooding event in September 2024, Storm Boris. We thereby demonstrate how the use of kilometer-scale resolution captures local-scale intensification with local granularity, including details missed by coarser models. This framework forms a core operational component of the EU's Destination Earth Climate Digital Twin, enabling scalable, high-resolution, circulation-constrained climate storylines for science and decision-making.

Plain Language Summary

This study introduces storyline simulations at kilometer-scale resolution using the IFS-FESOM climate model. This approach recreates recent extreme events, showing how they might change in different climate conditions while preserving fine local details through high resolution. We validate the system's ability to reproduce daily weather patterns and extreme events, comparing it to high-quality observations and field campaign data. The paper also highlights the storylines framework through examples, including the July 2019 heatwave and the 2024 Storm Boris in Europe. These storylines aim to create meaningful, tangible, and engaging insights into extreme weather events and how they may change with climate change.

1. Introduction

As climate change intensifies, extreme weather events are becoming more frequent and impactful, increasing the demand for climate information that is not only robust but also tangible and directly relevant for decision-making. Traditional climate projections provide essential long-term statistics but are less suited for addressing how specific extreme events are influenced by climate change. Moreover, even large climate models rarely contain true analogs of individual observed extremes, limiting event-specific impact assessments within classical simulation frameworks.

The broader concept of climate storylines (Shepherd & Lloyd, 2021; Shepherd et al., 2018) provides a framework for exploring physically plausible pathways of climate change and its impacts without relying on probabilistic

Antonio Sánchez-Benítez, Helge F. Goessling, Thomas Jung
Methodology: Amal John, Sebastian Beyer, Marylou Athanase, Antonio Sánchez-Benítez, Helge F. Goessling, Thomas Jung
Project administration: Sebastian Milinski, Irina Sandu, Thomas Jung
Resources: Amal John, Sebastian Beyer, Helge F. Goessling, Thomas Jung
Software: Amal John, Sebastian Beyer, Razvan Aguridan, Miguel Andrés-Martínez, Aina Gaya-Àvila, Suvarchal Kumar Cheedela, Philipp Geier, Rohit Ghosh, Ioan Hadade, Nikolay Koldunov, Xabier Pedruzo-Bagazgoitia, Thomas Rackow, Dmitry Sidorenko, Jan Streffing, Emanuele Vitali, Thomas Jung
Supervision: Amal John, Sebastian Beyer, Helge F. Goessling, Thomas Jung
Validation: Amal John, Sebastian Beyer, Marylou Athanase, Antonio Sánchez-Benítez, Helge F. Goessling, Thomas Jung
Visualization: Amal John, Sebastian Beyer, Akil Hossain, Matteo Nurisso
Writing – original draft: Amal John
Writing – review & editing: Amal John, Sebastian Beyer, Marylou Athanase, Antonio Sánchez-Benítez, Helge F. Goessling, Akil Hossain, Miguel Andrés-Martínez, Aina Gaya-Àvila, Suvarchal Kumar Cheedela, Rohit Ghosh, Nikolay Koldunov, Matteo Nurisso, Xabier Pedruzo-Bagazgoitia, Thomas Rackow, Irina Sandu, Dmitry Sidorenko, Thomas Jung

sampling. Building on this general concept, storyline simulations offer a more specific, simulation-based implementation, in which individual historical events are dynamically reconstructed and systematically replayed under alternative climate conditions (Hazeleger et al., 2015; Sánchez-Benítez et al., 2022; van Garderen et al., 2020; Wehrli et al., 2020). In the approach used here, only the large-scale atmospheric circulation is constrained using spectral nudging (see Feser & Shepherd, 2025, for an overview) to follow the observed evolution, while atmospheric thermodynamic and small-scale dynamical processes, as well as the coupled land surface, ocean, and sea ice, are allowed to evolve freely and respond consistently within the simulation. This coupled approach allows sea surface temperatures (SSTs) to respond consistently to local, nonlinear air-sea interactions, avoiding the inconsistencies of prescribed-SSTs methods often used in uncoupled storylines (Goosse et al., 2018; Holbrook et al., 2019; van Garderen & Mindlin, 2022; Wehrli et al., 2019).

Previous applications of storyline simulations have demonstrated their value for a wide range of extreme events, including heatwaves, marine heatwaves, cold outbreaks, and severe storms (Athanase, Sánchez-Benítez, Goessling, et al., 2024; Athanase, Sánchez-Benitez, Monfort, et al., 2024; de Vries et al., 2024; Klimiuk et al., 2024; León-FonFay et al., 2025; Sánchez-Benítez et al., 2022; Wehrli et al., 2020; Zhuo et al., 2025). However, these studies have relied almost exclusively on coarse-resolution global models (typically 50–100 km) or regional downscaling. While regional approaches, such as Pseudo Global Warming (Brogli et al., 2023), achieve high spatial resolution, they lack global spatio-temporal continuity and are susceptible to lateral boundary artifacts (Giorgi, 2019; Rummukainen, 2010). By extending the storyline framework to global kilometer-scale (km-scale) resolution in a coupled atmosphere–land–ocean–sea-ice model, the present work overcomes these limitations and enables event-specific climate change analyses at spatial scales directly relevant for impacts.

Global km-scale climate models resolve orography, coastlines, and land-surface heterogeneity far more realistically than conventional global models, improving the representation of precipitation, land-sea contrasts, and regional extremes. By explicitly resolving key physical processes, km-scale models reduce uncertainties associated with parameterized subgrid processes (Hohenegger et al., 2023; Jung et al., 2012; Kendon et al., 2019; Rackow et al., 2025; Slingo et al., 2022). Recent advances in high-performance computing have made such simulations feasible at the global scale, although their computational cost remains substantial, limiting the use of large ensembles (Segura et al., 2025). This places a premium on approaches that can extract physically meaningful information from relatively short simulations.

In this study, we introduce a global modeling framework that enables storyline simulations at km-scale resolution within a fully coupled climate model. The key novelties lie in (a) applying global km-scale resolution to ensure consistent representation of local-scale features and extremes, and (b) establishing the technical foundation for operational implementation within the Climate Change Adaptation Digital Twin developed under the European Commission's Destination Earth initiative (Doblas-Reyes et al., 2026; Hoffmann et al., 2023; Wedi et al., 2023, 2025). Together, these elements move storyline simulations toward an operational capability for climate science and climate services.

The remainder of the paper is organized as follows. We first describe the global km-scale model and experimental setup, followed by the nudging methodology. The present-day simulation is then evaluated against observations, including high-resolution data from the Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) field campaign. We subsequently present event-based storyline case studies and associated climate-change signals, before discussing implications, limitations, and future directions.

2. Technical Overview

2.1. Model Description

The km-scale storyline system described here utilizes the global climate model IFS-FESOM, which is based on ECMWF's Integrated Forecasting System (IFS) for the atmosphere, land and wave components, coupled to the Finite volumE Sea ice-Ocean Model (FESOM2) developed by the Alfred Wegener Institute (AWI). IFS utilizes a hydrostatic dynamical core with a two-time-level, semi-implicit, and semi-Lagrangian approach (Diamantakis & Váňa, 2022; Hortal, 2002; Temperton et al., 2001). FESOM2 is a global sea-ice ocean model that uses unstructured meshes composed of triangular cells whose size can vary smoothly to adapt the horizontal resolution spatially (Danilov et al., 2017; Scholz et al., 2019, 2021; Sidorenko et al., 2020). FESOM2 has been successfully

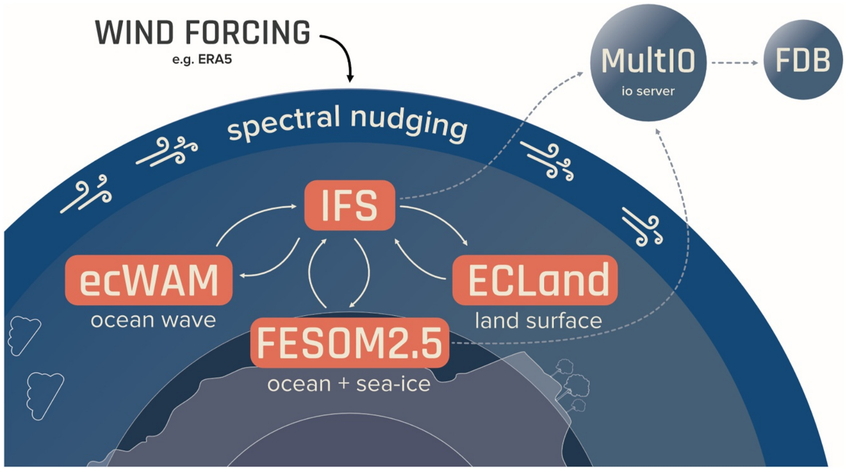

Figure 1. The coupled model consists of four main components: the Integrated Forecasting System as the atmospheric model, including ecWAM for ocean waves and ECLand for land-surface processes, and FESOM2 as the sea ice-ocean model. A detailed description of the coupled system and its components is provided by Rackow et al. (2025). Spectral nudging is applied to constrain the evolution of the large-scale atmospheric circulation with observations.

employed in previous coupled model setups (Rackow et al., 2025; Streffing et al., 2022). Figure 1 illustrates the different components of the IFS-FESOM configuration used here. Further details on IFS-FESOM can be found in Rackow et al. (2025).

We utilized a version of the IFS based on cycle 48r1, which has been operational at ECMWF since July 2023 (see ECMWF IFS documentation: https://www.ecmwf.int/en/publications/ifs-documentation). This configuration includes substantial modifications beyond the standard 48r1 cycle, including the implementation of an urban scheme and updated land-use and land-cover data sets, as documented by Rackow et al. (2025). Horizontally, IFS employs the TCo1279 grid (triangular–cubic–octahedral; see Malardel et al., 2016), corresponding to a nominal horizontal resolution of approximately 9 km. In the vertical, 137 model levels are used.

The sea ice-ocean component, FESOM2, employs 70 vertical levels and uses the unstructured NG5 mesh, which is designed to enable eddy-rich simulations over large parts of the global ocean (see Figure S10 in Supporting Information S2). The horizontal resolution of NG5 varies from approximately 12 km in the tropics, where the Rossby radius of deformation is large, to about 4.5 km in high latitudes and in dynamically active mid-latitude regions such as the Gulf Stream.

2.2. Spectral Nudging

We use spectral nudging to impose the observed evolution of the large-scale atmospheric circulation in the middle and upper troposphere, following the approach by Sánchez-Benítez et al. (2022). Also known as Newtonian relaxation, this method can be used for models to replicate specific large-scale circulation conditions in climate models by nudging toward reanalysis data or observations (Waldron et al., 1996; von Storch et al., 2000). The conceptual implementation of spectral nudging is illustrated in Figure S1 in Supporting Information S2. Here, we apply spectral nudging toward ERA5 reanalysis data obtained from the Copernicus Climate Change Service (C3S, Hersbach et al., 2023).

The nudging process involves relaxing the model state toward reanalysis data by adding a relaxation term to the prognostic equations (Miguez-Macho et al., 2004):

$$\frac{\partial X_n^m(\eta,t)}{\partial t} = F_n^m(\eta,t) + G_n^m(\eta)\left[X_n^{m(reana)}(\eta,t) - X_n^m(\eta,t)\right] \tag{1}$$

where m and n are the zonal and total wavenumbers, respectively; $X_n^m(\eta,t)$ and $X_n^{m(reana)}(\eta,t)$ represent the spectral coefficient of a meteorological variable at the vertical level η and time step t for the model and the reanalysis, respectively; $F_n^m(\eta, t)$ denotes the model forcing; and G_n^m represents the nudging coefficient, determining the strength of the nudging.

The impact of the nudging depends on which meteorological variables, wavenumbers, and vertical levels are constrained and at which strength (Omrani et al., 2015; Sánchez-Benítez et al., 2022; Schubert-Frisius et al., 2017; Separovic et al., 2015). Only vorticity and divergence are nudged in our experiments, allowing other atmospheric variables, as well as the land surface, the ocean, and sea ice, to evolve freely.

Here we employ triangular truncation of T60, which retains the total spectral wave numbers for vorticity and divergence up to 60. We limit the nudging to model levels between 100 and 700 hPa to allow the atmospheric planetary boundary layer and air-sea-ice coupling to develop as freely as possible. Sensitivity experiments indicated a small influence on the results when nudging was applied to the stratosphere or lower troposphere (see Sánchez-Benítez et al., 2022, their supplemental material). However, in the tropics, where the flow is driven by local convection and thermodynamics rather than the balanced, wave-driven dynamics that link the upper and lower levels in the extratropics, nudging restricted to above 700 hPa is less effective in constraining lower-tropospheric winds. This configuration is therefore optimized for extratropical dynamics, and results in the tropics should be interpreted with caution. Because large-scale winds are prescribed, part of the dynamically induced variability, particularly in the tropics and ocean, is constrained, and the resulting counterfactuals should be interpreted as thermodynamically adjusted scenarios rather than fully dynamically independent worlds. We also explored the sensitivity of our results to different nudging configurations (see Figure S2 in Supporting Information S2), providing insight into the robustness of our findings and the potential limitations of the approach (Radu et al., 2008).

Our approach applies a constant nudging strength across most of the nudged atmospheric levels, following established practice (Sánchez-Benítez et al., 2022; Schubert-Frisius et al., 2017; van Garderen et al., 2020; Wehrli et al., 2018). A sigmoid function is used to ensure a smooth vertical transition between nudged and non-nudged levels (see Sánchez-Benítez et al., 2022, Supporting Information S1). The nudging strength is characterized by an e-folding time, τ, with larger values corresponding to weaker relaxation toward reanalysis data. We adopt a relatively short e-folding time of 1 hr, which allows the model winds to closely follow the observed large-scale atmospheric evolution. At km-scale resolution (TCo1279), the combination of a 1-hr e-folding time with T60 spectral truncation represents a suitable compromise (see Figure S2 in Supporting Information S1), balancing effective large-scale constraint against the preservation of physically consistent variability at smaller spatial and temporal scales. This configuration ensures an effective representation of the observed weather, including extreme events (see Sections 3.1.3 and 3.2), while avoiding an undue constraint on the model's response to different background climate conditions.

Nudging is also commonly applied in grid-point space (e.g., Wehrli et al., 2021), in which model fields are relaxed toward the target data at every grid point and across all spatial scales. By contrast, the spectral nudging configuration employed here constrains only the large-scale circulation, while leaving small-scale processes at km-scale resolution unconstrained. Although the two approaches would theoretically converge if spectral nudging were configured to constrain all spatial wavenumbers, the spectral formulation offers the practical advantage of explicit and transparent scale selection.

2.3. Experimental Design

Our experimental design encompasses three distinct climates: the control (hereafter Cont) reflects conditions typical of 1950 and provides an analog for a cooler, early-industrial climate, historical (hereafter Hist), which represents present-day conditions (i.e., 2017–2024), and Tplus2.0K (hereafter Tp2K), which reflects a +2 K warmer than pre-industrial world (approx. 2040). Regarding the observed large-scale winds, all experiments are run for the period 2017–2024, with the same nudging input files (ERA5 see Section 2.2) and their respective forcings and initial conditions. A detailed overview of these experiments, including their respective periods, forcing scenarios, IFS initial conditions, and the FESOM2 spin-up, is provided in Table 1. These experiments represent different climate states, aligning with the concept of climate change storylines (Shepherd et al., 2018; van Garderen et al., 2020). The Cont experiment employs the historical forcing from the CMIP Phase 6, while the Hist and Tp2K simulations use forcing from the Shared Socioeconomic Pathway 3–7.0 (SSP370) scenario (O'Neill et al., 2016).

Table 1
Details for the Different Storyline Experiments

			Initial conditions	
Experiment	Description	Forcing	Atmosphere	Ocean
Cont	Control experiment with 1950 boundary conditions	CMIP6-historical (1950; fixed)	ERA5 (2017-01-01)	5 year standalone spinup (1945–1949)
Hist	Historical experiment with recorded boundary conditions	CMIP6-SSP3-7.0 (2017–2024; transient)	ERA5 (2017-01-01)	5 year standalone spinup (2012–2016)
Tp2K	Future experiment with boundary conditions from a 2 K warmer world	CMIP6-SSP3-7.0 (2040; fixed)	nextGEMS (2040-01-01)	nextGEMS (2040-01-01)

Note. The table provides a short description of the experiments along with the corresponding forcings and initial conditions used.

We implemented a spin-up procedure for all three experiments that balances computational constraints with the need for stable and comparable initial conditions. We conducted a stand-alone ocean spin-up for 5 years: from 2012 to 2016 for the Hist experiment and 1945 to 1949 for the Cont experiment. We have used the same atmospheric initial conditions derived from the ERA5 reanalysis data for the Cont and Hist. The Tp2K experiment, on the other hand, is branched off in the year 2040 from the coupled simulations started in 2020 in the context of the EU project NextGEMS (Rackow et al., 2025).

We acknowledge the inconsistency in ocean spin-up and the atmospheric setup for Tp2K compared to the Cont and Hist experiments. This discrepancy may lead to artifacts that affect the diagnosed climate change signal, including hemispheric differences in the background temperature fields where deep-ocean adjustment is important (e.g., in the Southern Ocean). This known shortcoming will be addressed when long coupled simulations with IFS-FESOM at the same resolution will become available in 2026. Our spin-up period is also shorter than the multi-decade or century-long spin-ups often used in climate modeling (e.g., Séférian et al., 2016). This represents a compromise necessitated by our experiments' high resolution and thus computationally intensive nature. Nonetheless, the spin-up is sufficient for faster coupled processes to allow ocean circulation and thermodynamic fields to adjust to the respective climate states.

To mitigate further potential initialization artifacts, especially the lack of initial atmospheric conditions for 1950 for the Cont experiments, and following the same approach as in Sánchez-Benítez et al. (2022), we incorporated an additional year of coupled model spin-up with nudging enabled. This allows for further adjustment of some of the more slowly responding parameters, such as soil moisture, sea-ice concentration (SIC), and SST, whereas some transient effects, particularly in the deep ocean, will remain. However, given our focus on relatively fast atmospheric and near-surface processes over a relatively short simulation period (2017–2024), we consider this limitation acceptable.

Figure 2 shows that each experiment represents a different climate: Cont has consistently lower zonal-mean temperatures than Hist, reflecting mid-20th century conditions, while Tp2K is notably warmer than Hist, reflecting the effects of continued global warming. Across most regions, the experiments align well with their intended climate states, with a smaller warming from Hist to Tp2K than from Cont to Hist. Globally averaged, the mean temperature increases by ≈1.3 K from Cont to Hist and by ≈2.3 K from Cont to Tp2K. The simulated warming of ≈1.3 K between Cont and Hist is broadly consistent with the observed global warming of ≈1 K estimated for the period between the 1950s and the early 2020s (Gulev et al., 2021; Lindsey & Dahlman, 2020; Masson-Delmotte et al., 2021). However, the inter-hemispheric differences in warming between Hist–Cont and Tp2K–Hist likely reflect both the differing forcing periods and residual differences in ocean spin-up and deep-ocean state.

To quantify the uncertainty associated with the unconstrained part of the simulations, we generated five ensemble members (E1–E5) for the period of 1 July to 31 December 2024. E1 represents the original simulation, where spectral nudging remained active throughout the period. The remaining four members were constructed by temporarily disabling spectral nudging starting on 1 July 2024 for increasing durations: 1 day (E2), 2 days (E3), 3 days (E4), and 4 days (E5). By re-enabling nudging after these varying intervals, we allowed small perturbations to develop and grow. This experimental ensemble approach enables a more robust assessment of events where

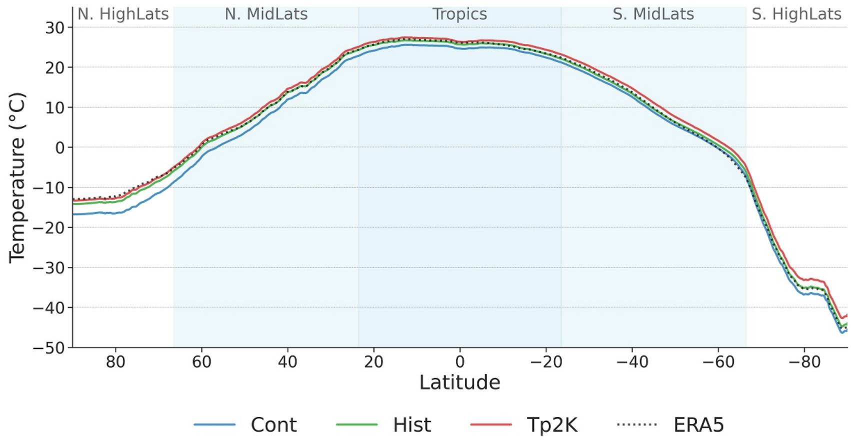

Figure 2. Zonal-mean 2 m-temperature (°C) averaged over 2018–2024 from three IFS–FESOM experiments, compared with ERA5 reanalysis. ERA5 is shown as a dotted line; IFS–FESOM results are shown for the control (Cont, blue), historical (Hist, green), and +2 K (Tp2K, red) simulations. Differences between the experiments reflect shifts in background climate conditions under different forcings, illustrating the model's sensitivity to imposed climate states, while part of the hemispheric contrast likely also reflects residual differences in ocean spin-up and deep-ocean adjustment.

small-scale variability matters, which is especially true for precipitation events. We plan to extend ensemble simulations to the full period in future work.

2.4. Computational Performance and Throughput

The experiments were conducted on Europe's first pre-exascale supercomputer, Large Unified Modern Infrastructure (LUMI), in Kajaani, Finland. We used the LUMI-C partition, consisting of dual-socket CPU nodes, each having two 64-core 3rd-generation AMD EPYC Milan CPUs. The core model was executed on 176 nodes with 16 MPI tasks per node and 8 OpenMP threads per task, amounting to a total of 2,816 MPI tasks. Additionally, 15 and 10 nodes were allocated for managing the I/O of the IFS and FESOM through multIO (Sarmany et al., 2024), respectively, resulting in a total number of nodes of 201.

This configuration yields an average throughput of 600 simulated days per day. The ocean model accounts for about 30% of the computational time. The hourly outputs are written on native, HEALPix (Gorski et al., 2005) and regular 0.25° lat-lon grids, using approximately 100 GB per day in compressed GRIB format. Autosubmit (Manubens-Gil et al., 2016) is employed as the workflow manager for all simulations to configure, execute, and monitor the experiments, to ensure reproducibility and traceability. Further details on the workflow can be found in Doblas-Reyes et al. (2026).

3. Results

3.1. Model Evaluation

Here, we evaluate the Hist (present-day, factual) simulation, focusing on its ability to capture key aspects of the climate system. We quantify how the Hist run manages to replicate observed temporal variability for different key parameters and regions. We then compare the model data with observations from the MOSAiC field campaign in the Arctic. Then, we assess the model's ability to represent two recent extreme events in Europe: a heatwave and an extreme precipitation event. Finally, we also compare the model's performance in capturing the fine-scale features and event-specific timing associated with different extreme events. Additional results from the model evaluation can be found in Figures S3–S10 in Supporting Information S2.

3.1.1. Temporal Variability

Figure 3 shows global maps of temporal correlation for daily 2 m-temperature (T2M), SST, total precipitation, and 500 hPa geopotential height (Z500), computed using ERA5 as a reference. The correlations, which determine the ability of the system to represent the observed temporal evolution of the parameters, were calculated after removing the mean seasonal cycle.

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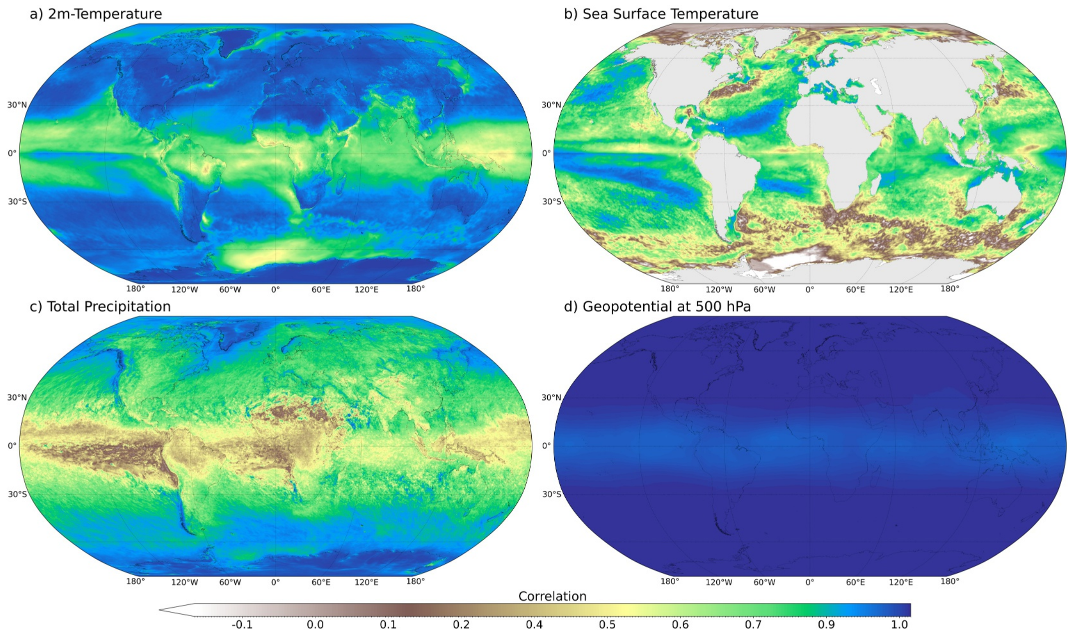

Figure 3. Temporal correlation between daily IFS-FESOM output from the Hist experiment (with spectral nudging) and ERA5 reanalysis for the period 2018–2024, after removal of the seasonal cycle. Correlation maps are shown for (a) 2 m-temperature, (b) sea surface temperature, (c) precipitation, and (d) 500 hPa geopotential height. A non-linear color scale is applied to highlight both very high and very low correlation values, with a consistent color scheme across all panels to facilitate visual comparison.

The correlation analysis for T2M (see Figure 3a) shows a very good agreement with ERA5 data, with values ranging from 0.8 to 1.0 across most regions with a clear latitudinal dependence. Evidently, the nudging technique performs best in the extra-tropics, where upper-tropospheric winds, including the jet streams, strongly determine near-surface weather. Not surprisingly, correlations for T2M are lower in the tropics, where the atmospheric flow is less barotropic and fast convective processes have a stronger influence on shaping the evolution of daily weather. However, albeit smaller than in the extratropics, the correlations of about 0.5–0.7 indicate that the model still captures a significant portion of the temperature variability in the tropics. Lower correlations in the Weddell and Lazarev seas are likely due to sea ice biases, where the presence or absence of sea ice strongly affects temperature and ocean-atmosphere heat exchange (Barber & Massom, 2007; Screen et al., 2018).

The correlation patterns for SST resemble those for T2M. However, values are generally lower with less pronounced latitudinal dependence. For most ocean areas, our results suggest that the model system is well suited for studying SST extremes, including marine heatwaves, across large parts of the world's ocean, consistent with previous findings using a coarser-resolution model (Athanase, Sánchez-Benítez, Goessling, et al., 2024). However, correlations drop in eddy-rich regions such as the Southern Ocean, the Gulf Stream, Kuroshio, and Agulhas. This decrease in correlation is expected given our model's high resolution (see Figure S10 in Supporting Information S2), allowing it to produce its own internal ocean weather that is not conditional on the atmospheric forcing. Furthermore, in the northern high latitudes, correlations for SST are particularly low where sea ice shields the ocean from direct atmospheric forcing and thereby suppresses ocean temperature variability (Vihma, 2014).

Correlations of 0.6 and higher for total precipitation (see Figure 3c) reveal that the Hist simulation performs well in representing day-to-day variability of precipitation in mid- and high-latitudes that is associated with large-scale synoptic systems (Jeuken et al., 1996). This is particularly true for orographically influenced precipitation. Notably, the South Asian Monsoon region shows temporal correlations above 0.5, indicating that interactions between large-scale circulation, orography, and land-sea thermal contrasts are effectively captured. This is in line with recent advancements in monsoon modeling (Wang et al., 2020). However, the model also shows reduced agreement across several key tropical regions, including the arid Sahara and Sahel, western continental margins, and the Amazon. In these regions, upper-tropospheric winds exert weaker control on surface weather, diminishing the influence of large-scale circulation patterns. Additional challenges arise from the representation of convective processes at sub-grid scales (Becker et al., 2021; Schiro et al., 2020), complex land–atmosphere interactions, vegetation patterns and associated feedbacks (Koster & Suarez, 2004), as well as the infrequent and intermittent nature of precipitation in arid environments. Together, these factors contribute to lower correlations and highlight the difficulty of constraining small-scale convective precipitation in lower latitudes by nudging large-scale winds, even in km-scale simulations (Prein et al., 2015).

Finally, temporal correlations for 500 hPa geopotential height exceeding 0.9 globally (see Figure 3d) confirm that the nudging closely constrains the weather patterns and planetary waves in the atmosphere. Correlations show a latitudinal gradient, with slightly lower values in the tropics and increasing to near 1 toward the poles, reflecting the varying strength of the large-scale dynamical control (Wallace & Gutzler, 1981).

3.1.2. Comparison With the MOSAiC Expedition Data

Next, we evaluate the factual simulation (Hist) against observations from the MOSAiC field campaign, following an approach similar to (Pithan et al., 2023). The comparison covers the period from October 2019 to October 2020 and includes dynamically active episodes such as warm-air intrusions from mid-latitudes in April 2020 (Rückert et al., 2023). This evaluation complements the global correlation analysis shown in Figure 3 by focusing on the model's performance along a well-observed trajectory under strong dynamical control. Because spectral nudging constrains only the large-scale winds in the free troposphere, the model closely follows the observed synoptic evolution along the MOSAiC drift trajectory while allowing the coupled ocean–ice–atmosphere system to respond freely at smaller scales.

Observed T2M data from the MOSAiC expedition for the period October 2019–October 2020 are compared with the Hist simulation in Figure 4a. The model reproduces the observed day-to-day temperature variability well, including rapid warming episodes associated with warm-air intrusions. This demonstrates the ability of spectral nudging to align the simulated and observed temporal evolution, enabling a meaningful assessment of remaining model deficiencies. The results confirm earlier findings based on coarser-resolution nudged simulations (Pithan et al., 2023), now extended to km-scale resolution.

Sea-ice concentration (SIC) along the MOSAiC drift trajectory is evaluated using satellite observations from AMSR-E and AMSR2 (89 GHz channels), averaged over radii of 6.25, 50, and 100 km around the Central Observatory (Kreienkamp et al., 2021; Krumpen et al., 2020; Melsheimer & Spreen, 2019). During winter, the Hist simulation agrees well with the observed near-complete ice cover (Figure 4b). At other times of the year, discrepancies emerge. In particular, the strong SIC decline observed between mid-April and mid-May—down to values of about 75 %—is known to result from an incorrect satellite retrieval (Krumpen et al., 2021) and is therefore not considered a model deficiency. By contrast, the delayed ice formation in October and November, reflected in lower-than-observed SIC, likely arises from the model's simplified treatment of sea ice: once ice is present, a constant ice thickness of 1.5 m is assumed when computing conductive heat fluxes. This reduces heat loss to the atmosphere during the formation phase and slows ice growth.

Sea-ice thickness observations from CryoSat-2 orbit-level 2P data and gridded CryoSat-2/SMOS multi-sensor products (October 2019–May 2020) (Hendricks et al., 2021; Krumpen et al., 2020, 2021) are compared with the model in Figure 4c. Observations are shown as mean values together with interdecile (10%–90 %) and interquartile (25%–75 %) ranges for 50 and 100 km search radii. The model maintains SIT values between about 1.5 and 3 m with relatively weak variability, falling outside the observed interdecile range for much of the period. In the central Arctic, where ice is typically thick, the neglect of snow-on-ice and the simplified thermodynamic treatment lead to overestimated conductive heat fluxes and excessive ice growth. These biases highlight a characteristic advantage of nudged coupled simulations: by minimizing mismatches in large-scale atmospheric evolution, remaining discrepancies can be more clearly attributed to limitations in model physics—here, sea-ice thermodynamics—than in free-running simulations (Trivedi et al., 2024).

Overall, the MOSAiC comparison demonstrates that the nudged IFS-FESOM configuration reproduces the observed temporal evolution of key atmospheric variables while exposing specific deficiencies in sea-ice representation. These shortcomings are expected to be alleviated in forthcoming model versions that include improved sea-ice thermodynamics and coupling.

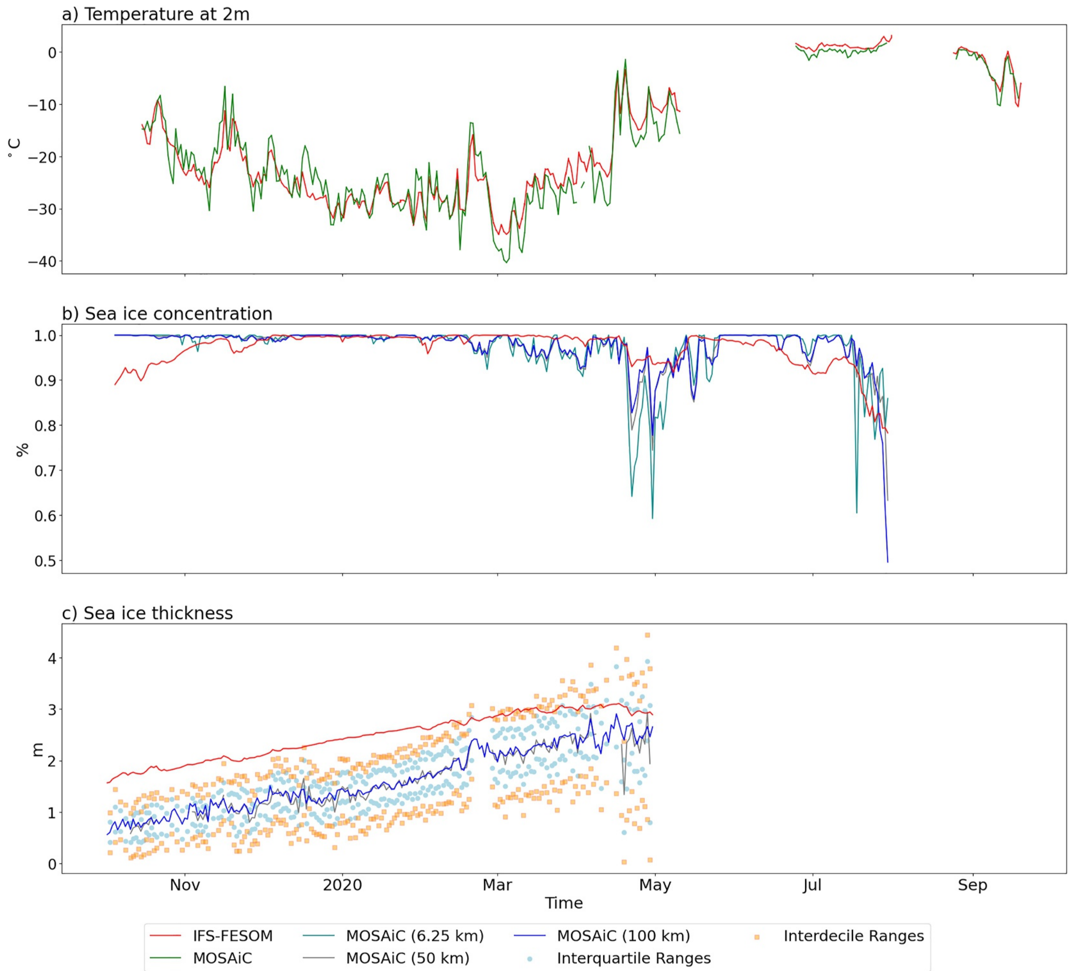

Figure 4. Daily time series along the MOSAiC drift trajectory at the Central Observatory location. Shown are (a) 2 m-temperature (°C) from October 2019 to October 2020, (b) sea-ice concentration (%) from October 2019 to July 2020 averaged over radii of 6.25 km (dark cyan), 50 km (gray), and 100 km (blue), and (c) sea-ice thickness (SIT) (m) from October 2019 to May 2020. Sea-ice thickness observations are shown as the mean together with interdecile (10%–90%) and interquartile (25%–75%) ranges. Red curves indicate the corresponding values from the spectrally nudged IFS–FESOM Hist simulation. Time-series lengths vary by variable due to observational data gaps.

3.1.3. Capturing Extreme Events

The ability of climate models to accurately capture extreme weather events is crucial for our understanding and capacity to predict the impacts of climate change. Here, we determine the capability of our km-scale configuration to reproduce small-scale features of extreme events based on two recent events in Europe: a heatwave and a heavy precipitation event. We compare our km-scale nudged configuration with results from a lower-resolution counterpart (≈50–100 km) using the AWI-CM1 model (Sánchez-Benítez et al., 2022) using ERA5 reanalysis data along with a merged gridded observational data set E-OBS (Cornes et al., 2018; Figure 5c and MSWEP Beck et al., 2017; Figure 5g).

For the European heatwave on 25 July 2019 (Figure 5, upper panels, Sánchez-Benítez et al., 2022; Vautard et al., 2020), AWI-CM1 and IFS-FESOM capture the general pattern of the heatwave. The km-scale simulation with IFS-FESOM, however, depicts a much more detailed picture, demonstrating local granularity, with orographic features, particularly in the mountainous regions of the Alps as well as in the urban areas, standing out. Notably, IFS-FESOM captures the highest temperatures over Paris, where the city experienced its highest-ever recorded temperature that day. This added realism is made possible by the combination of km-scale resolution and the explicit inclusion of an urban scheme, both of which are absent in previous lower-resolution

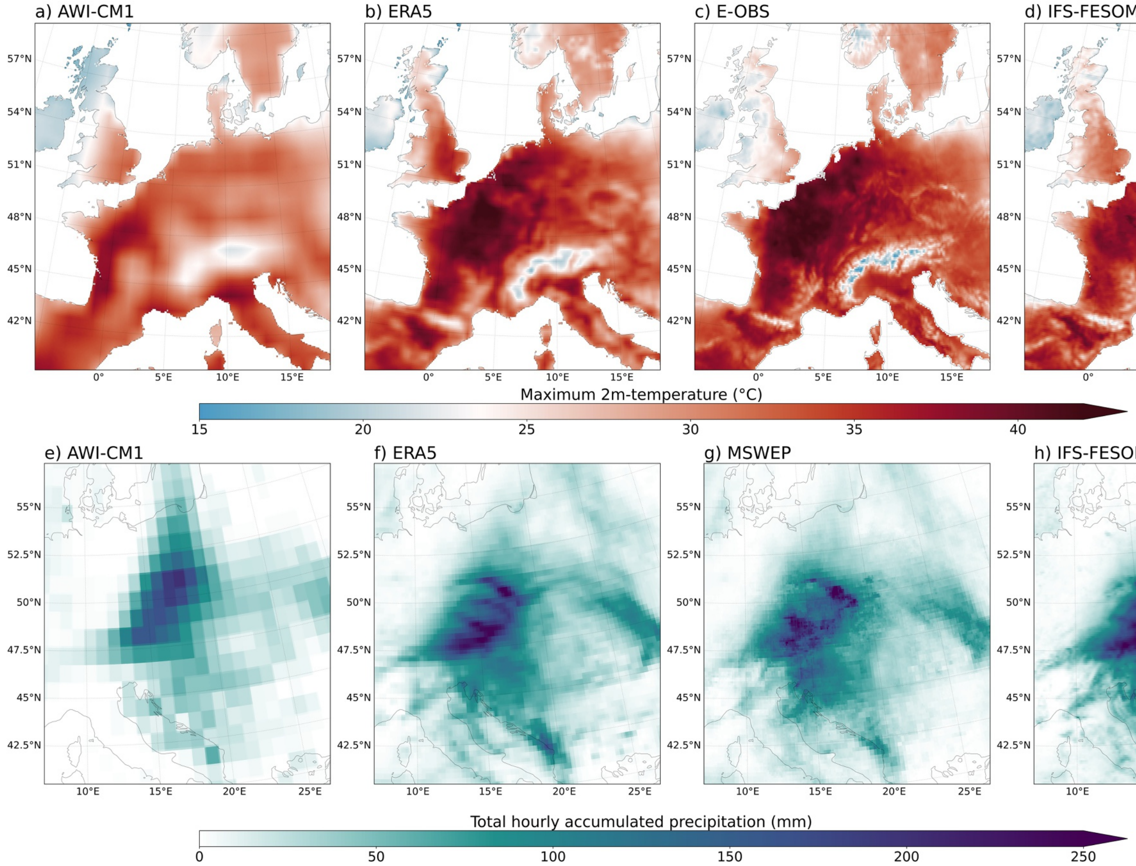

Figure 5. Comparison of simulated and observed extreme events using km-scale IFS-FESOM simulations and the coarser-resolution AWI-CM1 model. Panels (a–d) show the maximum 2 m-temperature (°C) during the peak of the 25 July 2019 European heatwave; panels (e–h) show 5-day accumulated precipitation (mm) for Storm Boris (12–16 September 2024). Observations and reanalysis data include E-OBS, MSWEP, and ERA5. Relative to AWI-CM1, IFS-FESOM shows closer agreement with these reference data sets, highlighting the added value of km-scale resolution for reproducing observed extremes.

configurations. Together, they enable a more accurate representation of land-atmosphere interactions, including urban-atmosphere exchanges and local-scale circulations like mountain-valley breezes and urban heat island effects, which aid in the development and persistence of heatwaves at the local scale (Heaviside et al., 2017; Knist, 2018). In contrast, the coarser resolution AWI-CM1, while capturing large-scale aspects of the heatwave, lacks the detailed topographic influences on temperature distribution, which are crucial for local-scale impact assessments (Hamdi et al., 2020; Maraun et al., 2017; Rangwala & Miller, 2012). These clear differences between the two models underscore the importance of high-resolution simulations with an urban scheme to capture the richness of heatwaves across spatial scales.

A similar analysis for Storm Boris (Athanase, Sanchez-Benitez, Monfort, et al., 2024), which led to widespread flooding in parts of Europe during the period 12–16 September 2024 (Figure 5, bottom panel), further supports that the km-scale configuration adds important and realistic small-scale structure seen from observations. In fact, IFS-FESOM reproduces extreme precipitation with a high spatial detail and timing that is in close agreement with MSWEP and ERA5. In particular, it captures regional variability and localized precipitation maxima across catchment areas, which are important for impact assessments. In contrast, the coarse-resolution AWI-CM1 simulation falls short of capturing the observed spatial pattern and magnitude of the precipitation (peak values of ≈200 mm compared to values greater than ≈225 mm in ERA5, MSWEP, and IFS-FESOM), though it reasonably replicates the affected area's geographic extent and the exceptional nature of the event (Athanase, Sánchez-Benitez, Monfort, et al., 2024). This reflects a common limitation of coarser resolution models that often underestimate extreme precipitation intensities and misrepresent the spatial distribution of rainfall, as documented in previous studies (Fosser et al., 2015; Kendon et al., 2017; Prein et al., 2015). The higher accuracy of our km-scale simulation in capturing the event is in agreement with prior research that has emphasized the role of high-resolution modeling for accurately depicting extreme precipitation (Ban et al., 2014; Becker et al., 2021; Kendon et al., 2012; Ludwig et al., 2023).

3.1.4. Model Performance Metrics for Extremes

To further evaluate IFS-FESOM's ability to simulate temperature and precipitation extremes in our nudged configuration, we compare its performance to ERA5 reanalysis data for the period 2018–2024 using a standard suite of temperature- and precipitation-related extreme indices, defined by the Expert Team on Climate Change Detection and Indices (ETCCDI, see details in Table S1 in Supporting Information S2; Sillmann et al., 2013; Zhang et al., 2011). These indices capture different aspects of extremes, including their intensity, frequency, and duration. They are typically averaged over multiple years to obtain a climatological mean, which is then used to compute differences or spatial correlations between data sets, such as between free-running climate models and observations. In contrast, here we compare indices for each month individually. This approach would not be suitable for a free-running model, where inter-annual variability is unrelated to the observed changes; in our case, however, this approach is feasible since temporal coherency with observations is imposed by the wind nudging. Therefore, rather than focusing solely on how well the model reproduces the climatology of key variables, our approach assesses its ability to capture recent observed extremes.

To quantify skill, we use two complementary metrics: the root mean square error (RMSE) and spatial pattern correlation. The (area-weighted) spatial and temporal mean RMSE quantifies the average magnitude of deviations between the model simulations and ERA5. The temporal mean spatial pattern correlation assesses how well the model replicates the spatial patterns of extreme events as they occurred during the period 2018–2024. Using both metrics allows us to differentiate between errors in magnitude and in spatial organization. As a benchmark, we also include results from the coarser-resolution nudged simulations with AWI-CM1, allowing us to isolate the added value of km-scale resolution while keeping the nudging approach consistent. The indices were first computed on the native grid of each data set and subsequently remapped to the IFS-FESOM grid for comparison.

Figure 6a displays the RMSE values, where higher values (darker blue shades) indicate greater discrepancies from ERA5, while lower values (lighter shades) signify better agreement. These values are not comparable across indices due to differing units and do not directly reflect simulation quality. However, the plot is best interpreted by examining relative differences between the two models for each index and region. At the global scale (GLB) and for all regions, both the coarse and km-scale models generally exhibit low RMSE, with an overall light yellow and green shading indicating that both models capture the observed values of precipitation and temperature indices. However, regional and index-specific differences are apparent, particularly for the monthly maximum one-day precipitation (rx1day) index in the tropics, and to a lesser extent in the northern and southern mid-latitudes. Despite these regional challenges with rx1day, IFS-FESOM consistently shows lower RMSE values than AWI-CM1 across nearly all indices and regions. For instance, in the tropics, IFS-FESOM's RMSE for rx1day is $\approx$6.5 mm, whereas AWI-CM1 displays a larger RMSE of $\approx$8.5 mm. This difference signifies that, in the tropics, IFS-FESOM simulates the monthly daily maximum rainfall on average 2 mm closer to ERA5 values than AWI-CM1.

Figure 6b illustrates the spatial correlation values, with higher correlations (lighter shades) indicating strong agreement and lower correlations (darker shades) reflecting worse representations of spatial patterns. While both models exhibit high correlation values across most indices and regions, IFS-FESOM consistently captures the spatial patterns better than AWI-CM1. This advantage is particularly evident for precipitation-related indices. Even in challenging regions like the Tropics, IFS-FESOM maintains high correlation values for precipitation extremes indices (typically >0.75), indicating its effectiveness in representing convective variability more accurately. When it comes to temperature extremes, which typically occur on larger scales, AWI-CM1 performs almost good as IFS-FESOM, at least when integrated over larger areas.

3.2. km-Scale Climate Change Storylines for Extreme Events

In the following, we present climate storylines for two extreme events constructed from our km-scale simulations with nudging under different climate conditions. The focus here is on demonstrating the storyline framework and its capabilities, while a detailed analysis of the underlying physical drivers for the changes is left for future work.

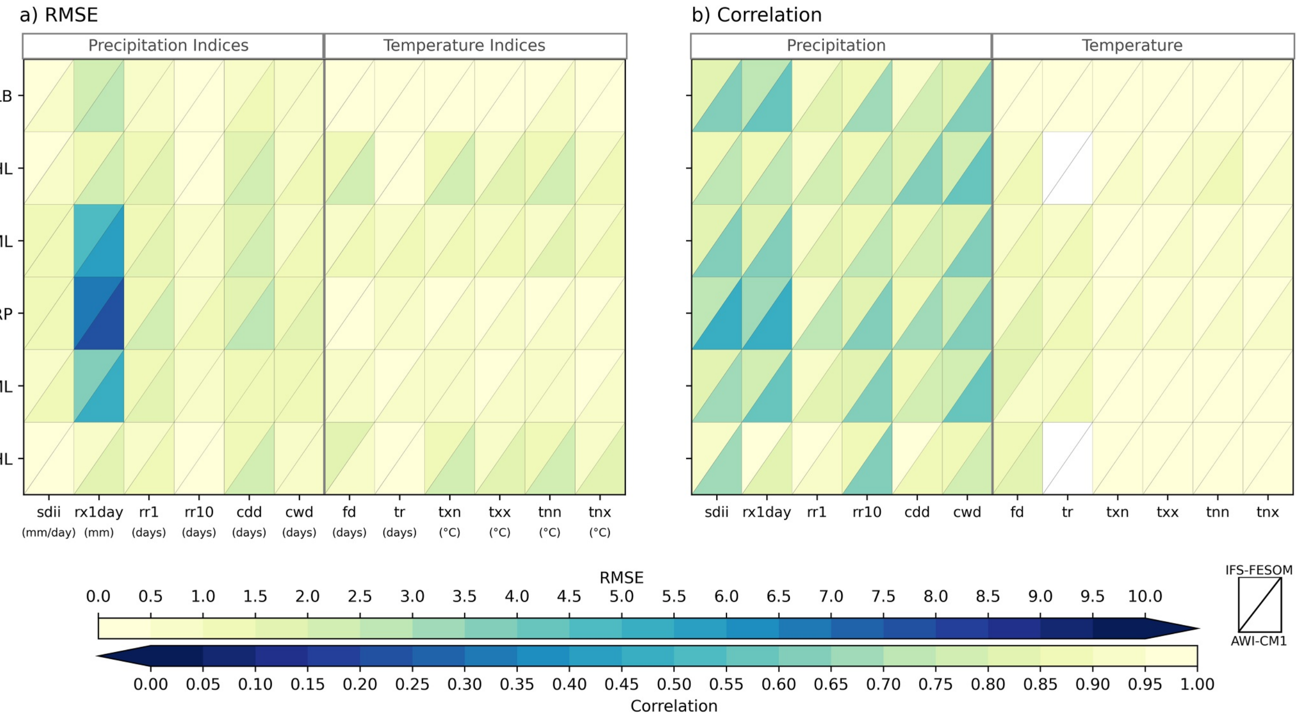

Figure 6. Root mean square error (RMSE; unit varies by index) and temporal pattern correlation for monthly temperature and precipitation extreme indices, based on (Sillmann et al., 2013; Zhang et al., 2011), from nudged IFS-FESOM and AWI-CM1 simulations (2018–2024) compared to ERA5. RMSE is computed as an area-weighted spatiotemporal average, while correlation reflects the mean pattern agreement with ERA5. Rows correspond to regions: Global (GLB), Northern/Southern High and Mid-Latitudes (NHL/SHL, NML/SML), and Tropics (TRP). Columns indicate 12 Expert Team on Climate Change Detection and Indices indices, grouped by precipitation and temperature categories. Each grid cell is split diagonally: IFS-FESOM (upper left triangle) and AWI-CM1 (lower right triangle). No color (white) indicates insufficient data available for the index. Definitions of indices and regions are provided in Tables S1 and S2 in Supporting Information S2.

Figure 7 presents the maximum T2M over France during the peak of the 25 July 2019 heatwave, when temperatures in Paris reached 42.6°C. The maps show spatial distributions of daily maximum temperatures for the Hist, Cont, and Tp2K experiments. The simulations show a clear increase in maximum temperatures from the Cont to the Tp2K experiments, along with a notable increase in the extent of the high-temperature (>40°C) areas in Tp2K as indicated by the yellow contour lines. To assess local intensity and its changes more closely, Figure 7d shows the hourly areal maximum temperature within the box covering Paris (see the black boxes in Figures 7a–7c). The Cont experiment does not reach 40°C at any time, with the maximum recorded temperature being 38.4°C. However, the Hist and Tp2K experiments simulate temperatures exceeding 40°C, peaking at 40.9°C and 41.3°C, respectively. At the time of writing, only a single member was available for each of the experiments in 2019; extending the ensembles produced for autumn 2024 (see below) to cover the full simulation period and hence allow uncertainty quantification will be a priority for the near future. However, the previous work by Sánchez-Benítez et al. (2022) demonstrates that for heatwaves, even a single ensemble member is sufficient to capture some of the salient features of change.

Both Hist and Tp2K experiments highlight how anthropogenic warming could intensify extreme heat events, projecting longer durations of temperatures exceeding 40°C in the Paris region and a potential temperature rise of ≈2.5°C–3°C compared to the past.

This approach could be further enhanced by using these storyline scenarios to drive impact models, enabling a more comprehensive understanding of factors relevant to impacts and clearer projections of future extremes at the local scale. The fine spatial detail of IFS-FESOM allows it to capture temperature variations in complex terrains, such as urban and mountainous regions, which is particularly relevant for analyzing emerging urban heat islands and their development under climate change (Nuruzzaman, 2015; Parker, 2010).

Next, we consider Storm Boris that brought widespread flooding in parts of Europe during the period 12–16 September 2024. For this event, a five-member ensemble has been generated (E1–E5) for each climate

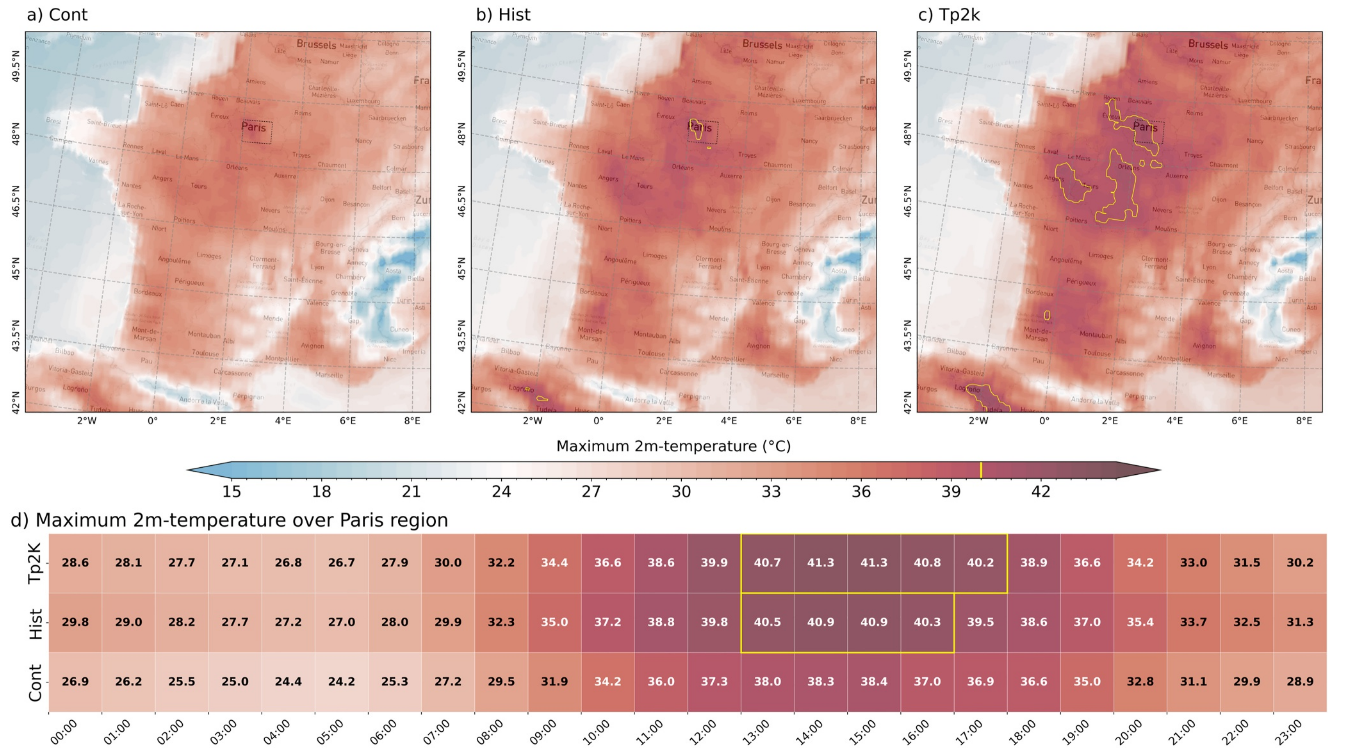

Figure 7. Maximum 2 m-temperature (°C) during the 25 July 2019 heatwave over France from the different IFS-FESOM storyline experiments. Panels (a–c) show spatial fields for the peak day; yellow contours indicate areas exceeding 40°C. Panel (d) shows the peak hourly 2 m-temperature within the Paris region (spatial maximum in the area 48.5°N–49°N, 2.0°E–3.0°E; outlined by a black dotted box in maps). Yellow-shaded intervals mark periods when temperatures exceed 40°C. The figure illustrates how the intensity and timing of the heatwave vary across different climate storylines.

background. Compared to the individual ensemble members, the ensemble mean exhibits reduced noise while retaining a fine-scale climate change signal (Figures 8a–8c). There is an expansion of approximately 19.4% of the area affected by extreme rainfall (>100 mm) between Cont and Hist experiments, followed by a more moderate 3.5% additional increase in Tp2K. These results are consistent with the findings by Athanase, Sánchez-Benitez, Monfort, et al. (2024) with the coarse-resolution AWI-CM1 system, who reported a climate-driven expansion of the extreme rainfall area for storm Boris from pre-industrial to present-day climate, and a smaller projected continued growth for a globally +4 K warmer climate. While low-resolution and km-scale models tell a consistent story about the general climate change signal, the km-scale model's improved spatial resolution allows for a more detailed representation of the rainfall changes, especially associated with orography. The small-scale structures are of critical importance to quantify local impacts of climate change, such as stronger landslides and flooding (see Figure 5e).

Integrated over the region defined by the red polygon in Figure 8 (top panels), our analysis reveals a 10% increase in total accumulated precipitation from the Cont (224 mm) to Hist (247 mm) simulations. For Tp2K (256 mm), precipitation increases by a further 3.5%. These results are qualitatively comparable to those reported by Athanase, Sánchez-Benitez, Monfort, et al. (2024), where the largest increase in extreme rainfall was attributed to climate change over the historical period (present vs. preindustrial), with only a moderate increase projected under stronger future warming. However, quantitatively, our results indicate a stronger intensification of the extreme precipitation, in particular when taking into account that Athanase, Sánchez-Benitez, Monfort, et al. (2024) considered a much stronger warming of +4 K instead of the +2 K warming considered here. One plausible explanation is that km-scale modeling enhances the sensitivity to moderate warming by explicitly resolving mesoscale circulation adjustments and orographic effects, which can modify the climate-change signal when aggregated over larger spatial domains (Abbott et al., 2020).

The temporal evolution of rainfall rates over the course of the event (Figure 8d) is highly coherent between the three simulations, with increased rainfall in warmer climates, particularly during the phase of peak intensity on the

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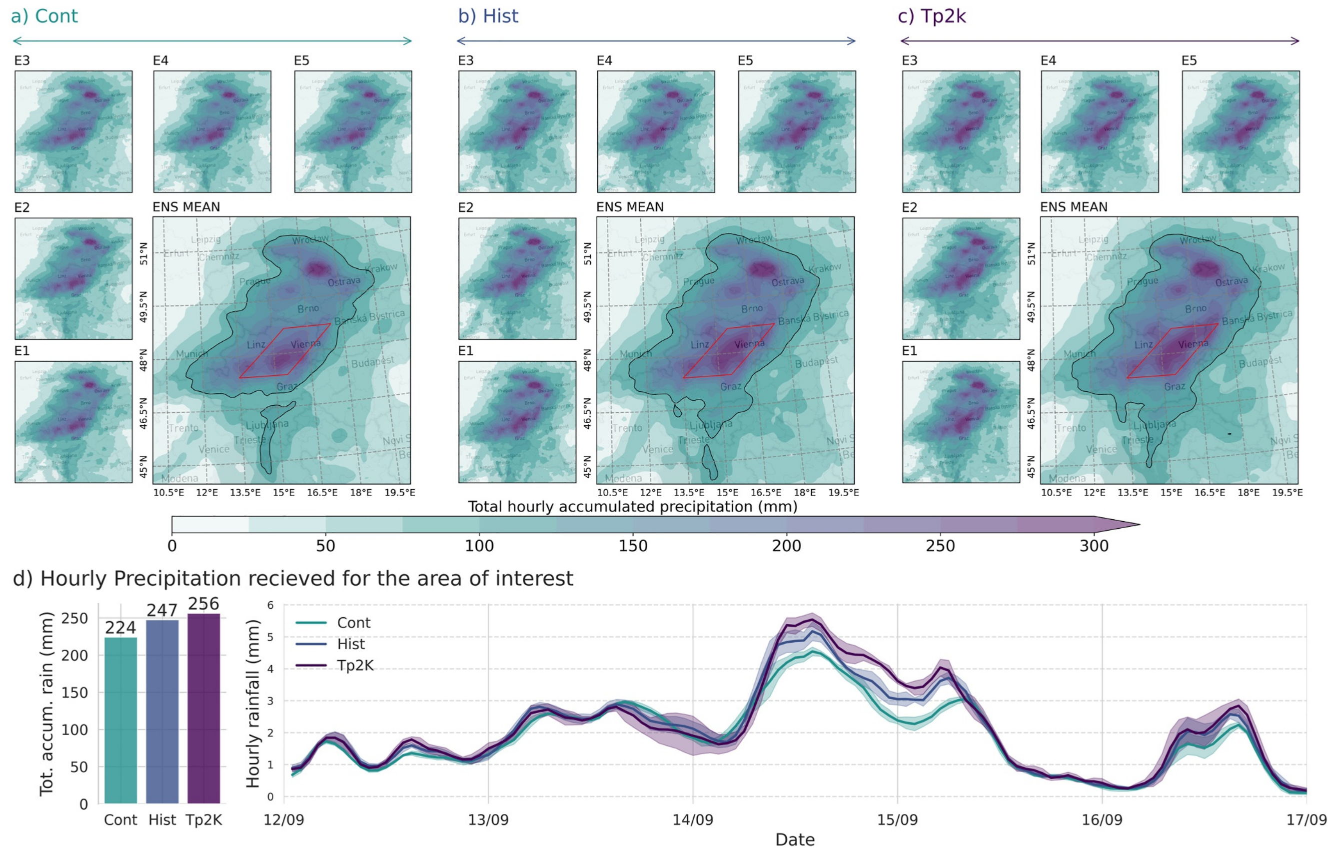

Figure 8. Simulated cumulative precipitation (mm) during Storm Boris (12–16 September 2024) from different IFS-FESOM experiments with nudging. The top panels show spatial distributions: E1–E5 represent individual ensemble members; "ENS MEAN" panels show the ensemble mean for each experiment. Black contours in the ensemble mean maps mark areas receiving more than 100 mm of rainfall. The bottom charts display total event precipitation per experiment (bars) and hourly precipitation time series within the area outlined by the red polygon in the top panels. The comparison illustrates how the same extreme rainfall event (Hist) might have unfolded in a past (Cont) or future (Tp2K) climate.

14 and 15 September. Other differences between the simulations regarding amplitude and timing of the rainfall are mostly small compared to the ensemble spread (indicated by shading around the curves in Figure 8d), suggesting that the impact of climate change on precipitation and its signal-to-noise ratio can be strongly flow-dependent.

3.3. Annual Mean Thermodynamic Climate Change Signals

In addition to analyses of short-term events, the circulation-constrained framework introduced here can be extended to longer temporal aggregates, including entire years. This enables climate storylines that capture the cumulative expression of weather events over a year. Annual-scale storylines provide a complementary perspective on climate variability by addressing how a specific year unfolds under the observed large-scale atmospheric circulation, and how that same year would differ under alternative background climate conditions. Such an approach could be particularly relevant for the interpretation of recent years characterized by exceptional climate conditions, including record-breaking global or regional mean temperatures (Goessling et al., 2025). Rather than estimating long-term mean climate change, annual storylines aim to contextualize individual years by separating mostly thermodynamic effects associated with background climate change from the influence of the specific atmospheric circulation realized in that year.

To illustrate this, we analyze annual-mean 2 m temperature (T2M) differences between the +2 K warmer (Tp2K) and present-day (Hist) simulations for recent individual years (2022, 2023, and 2024). These two climate states associate with distinct ocean circulation states, including a slowdown of the Atlantic Meridional Overturning Circulation (AMOC) in the Tp2K simulation. As a result, the annual storylines capture the combined thermodynamic response to warming and the imprint of ocean circulation changes, including a North Atlantic warming-hole-like fingerprint, while the large-scale atmospheric circulation is nudged to observations. By assessing annual-mean differences in nudged simulations, one can identify deviations attributable to non-circulation causes (or model error), as is often required for assessing the state of the climate (Blunden et al., 2025).

In the nudged simulations, annual-mean temperature differences exhibit spatially consistent warming patterns across different years (Figures 9a–9c), indicating that the imposed climate change forcing dominates over residual internal variability at the annual scale when the large-scale circulation is constrained. The reproducibility of these spatial patterns is quantified using pairwise pattern correlations between individual years (e.g., Tp2K–Hist in 2024 vs. 2023), which are high in the nudged simulations ($\rho = 0.7$–0.9; supplemental Table S3 in Supporting Information S2).

To contrast these results with an unconstrained framework, we construct an analogous diagnostic from the free-running nextGEMS simulations by differencing individual year pairs (2038 minus 2020, 2039 minus 2021, and 2040 minus 2022), representing an indicative contrast between a ≈2 K warmer future and present-day conditions (see Figures S11a–S11c in Supporting Information S1). In these simulations, single-year differences exhibit substantially weaker spatial coherence and lower pattern correlations ($\rho = 0.4$–0.6; Table S3 in Supporting Information S2) compared to the nudged runs. This does not imply an absence of a forced climate-change signal, but rather reflects the dominance of stochastic, dynamically driven circulation variability, which masks the thermodynamic response when individual years are considered in isolation. While robust signals in such free-running frameworks typically emerge only after multi-year or ensemble averaging, the circulation-constrained approach effectively filters this dynamical noise. This allows a robust signal to be extracted from only a single year of data, providing a computationally efficient alternative for isolating the forced thermodynamic response.

The primary advantage of the nudged simulations is therefore a significantly higher signal-to-noise ratio for short temporal aggregates. By suppressing a large fraction of chaotic atmospheric variability, the annual thermodynamic signals associated with background warming and coupled ocean responses emerge clearly without the need for large ensembles. This enables a physically interpretable decomposition of specific annual anomalies, providing a method to contextualize how a particular year, such as the record-breaking 2023 or 2024, would have unfolded under different background climate conditions while preserving the observed sequence of weather events.

The clarity of this signal allows for the identification of key spatial features, such as the distinct cooling signal in the North Atlantic. This "warming hole" is consistent with the documented fingerprint of a weakened AMOC (Menary & Wood, 2018). Similarly, the simulations clearly capture the expected land–sea warming contrast, with enhanced warming over continental interiors (Sutton et al., 2007). Finally, a cold anomaly observed north of the warming hole (Figures 9a–9c) likely reflects residual model drift associated with differing spin-up procedures between the Tp2K and Hist experiments, demonstrating that the framework is sensitive enough to isolate subtle model-specific biases even at the annual scale.

4. Discussion

This study presents, to our knowledge, the first global modeling framework that enables km-scale, event-specific climate change storylines within a fully coupled Earth system model. By combining global km-scale resolution with scale-selective spectral nudging, the framework makes it possible to reconstruct specific observed weather events, including extremes, under different climate conditions while preserving local-scale processes and coupled atmosphere–land–ocean–sea-ice interactions. This represents an advance over previous approaches, which have typically been limited to coarser resolutions, regional domains, or atmosphere-only configurations (de Vries et al., 2024; Feser & Shepherd, 2025; Hazeleger et al., 2015; Klimiuk et al., 2024; Sánchez-Benítez et al., 2022; van Garderen et al., 2020; Wehrli et al., 2019).

A key strength of the approach is the conditioning on the observed large-scale atmospheric circulation. By constraining the large-scale flow while allowing other processes to evolve freely, the simulations achieve a higher signal-to-noise ratio in climate signals than free-running climate simulations. This enables physically interpretable diagnostics of climate change signals for individual extreme events and even for entire years, without reliance on long-term averaging or large ensembles. For km-scale global models, where computational constraints limit the feasibility of large ensembles, this represents a practical advantage rather than the primary

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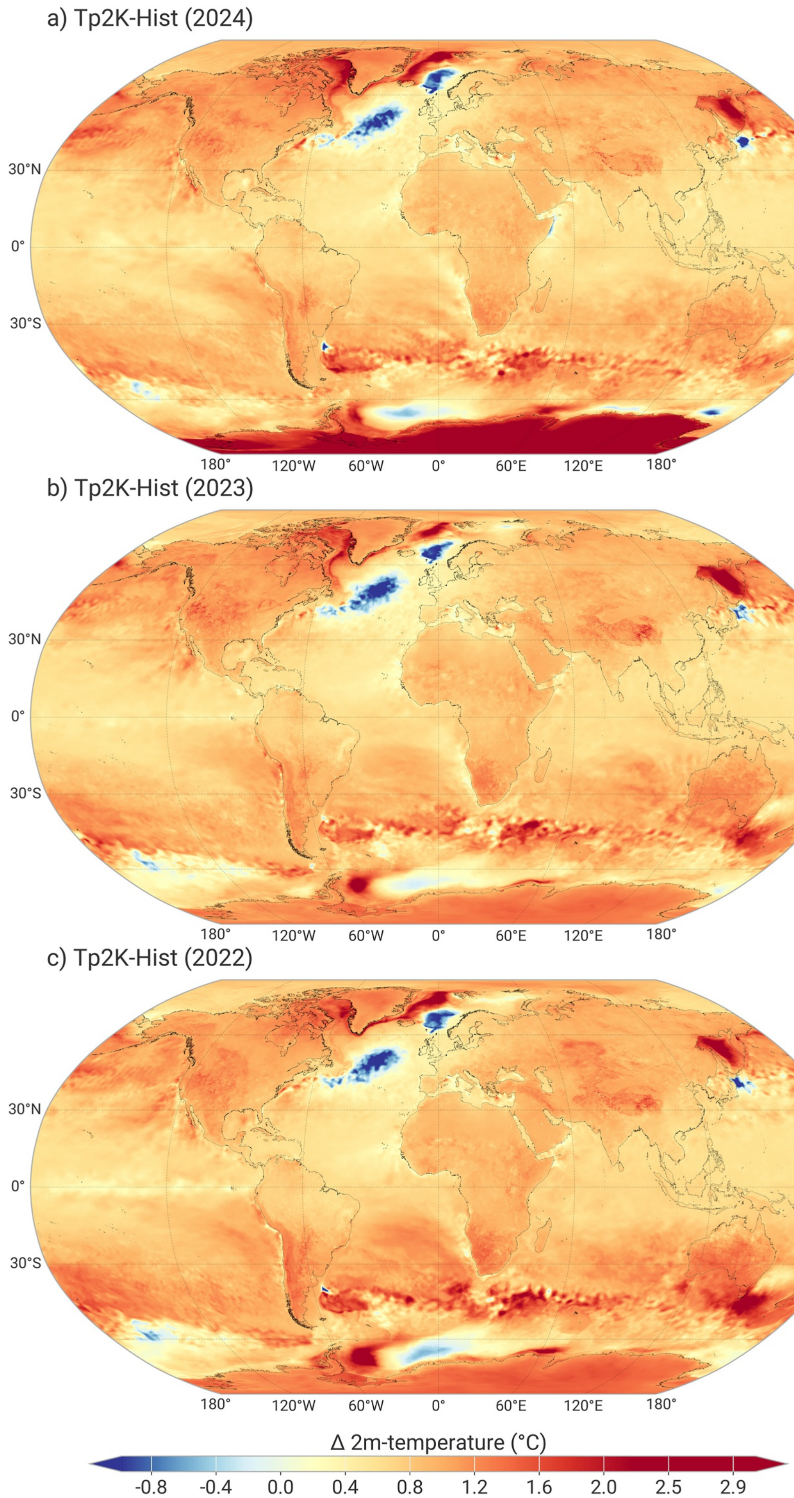

Figure 9. Maps show annual mean 2 m-temperature differences (°C) between the +2 K warmer world (Tp2K) and present-day conditions (Hist) for the years (a) 2024, (b) 2023, and (c) 2022. The comparison highlights that nudging yields spatially coherent thermodynamic signals at the annual scale. For a comparison with the single-year differences from unconstrained free-running simulations, see Figure S11 in Supporting Information S1.

motivation of the framework. As demonstrated here, this property extends naturally from event-scale to annual-scale storylines, providing a useful basis for interpreting exceptional years, including those associated with record-breaking temperatures.

At the same time, the interpretation of the results must account for the deliberate conditioning inherent in the approach. The simulations do not represent independent realizations of the climate system, but thermodynamically adjusted versions of specific dynamical situations. The constraint is most effective in the extratropics, while results in the tropics, particularly for precipitation, should be interpreted with appropriate caution. Residual variability also remains, especially in regions influenced by oceanic mesoscale processes that are less tightly constrained by atmospheric nudging.

Several known limitations of the current configuration should also be acknowledged. Differences in initialization and spin-up between the present-day and +2 K simulations can introduce residual inconsistencies, particularly in the ocean and sea-ice state, which may influence regional signals. These effects are expected to be reduced in future applications based on longer, internally consistent coupled simulations, such as those being produced within the EU-funded EERIE project. In addition, aspects of the sea-ice representation remain simplified in the present setup, which can affect ice growth, thickness, and associated feedbacks. These limitations are well understood and are being addressed in forthcoming updates of the operational system. Importantly, they do not undermine the core capability demonstrated here, namely the generation of global, km-scale, event-specific climate storylines, but rather point to clear pathways for further improving physical realism.

Beyond the scientific insights presented here, the framework provides a direct pathway toward impact-oriented applications. Km-scale, event-specific storylines are particularly well suited for impact sectors such as wind energy, wildfire risk, hydrological extremes including floods and droughts, extreme precipitation, and cryosphere-related applications. The high spatial and temporal resolution, combined with global consistency, allows sector-specific indicators to be derived without intermediate downscaling, while preserving the spatial coherence and timing of extreme events. Impact models driven by the km-scale global simulations are currently being implemented, enabling end-to-end, physically consistent analyses from large-scale circulation down to local impacts.

Looking ahead, the storyline framework presented here provides a flexible basis for further extensions. Future work will expand the range of background climate states considered, including higher warming levels, and increase ensemble coverage where feasible. Ongoing developments also aim to strengthen the coupling to impact models across sectors, enabling more comprehensive assessments of climate-change impacts under event-specific storylines. In parallel, efforts are underway to move toward more automated, near-real-time capabilities, allowing the rapid reconstruction and analysis of extreme events as they unfold (e.g., Athanase, Sánchez-Benitez, Monfort, et al., 2024). Together, these developments point toward a system that not only supports retrospective analysis but also enables timely, climate-informed narratives of current and emerging extremes.

The km-scale storyline system described here forms one component of the operational climate projection system developed as part of the Climate Digital Twin of the European Commission's Destination Earth Initiative (Doblas-Reyes et al., 2026; Hoffmann et al., 2023; Wedi et al., 2023, 2025). In this context, the framework is designed not only as a research capability but as an operational system that supports scalable, high-resolution, circulation-constrained climate projections. This operational embedding ensures continuity between scientific development and downstream applications, while allowing the system to evolve as model physics, computational resources, and data infrastructures advance (Bauer et al., 2021; Stevens et al., 2024).

Overall, global km-scale climate storylines offer a powerful bridge between weather-scale extremes and climate-scale assessments. By linking specific events and years to alternative climate backgrounds in a consistent way, the framework provides a robust foundation for both scientific analysis and emerging climate services, including applications within digital twin environments.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

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Availability Statement

Data from the IFS-FESOM storyline runs are available through the DestinE Platform with an upgraded access: https://platform.destine.eu/; technical examples and code for downloading this data via the Polytope API can be found at https://github.com/destination-earth-digital-twins/polytope-examples/tree/main. ERA5 reanalysis data used in this study can be downloaded from (Hersbach et al., 2023): https://doi.org/10.24381/cds.adbb2d47. E-OBS can be downloaded from (Cornes et al., 2018): https://doi.org/10.24381/cds.151d3ec6. The MSWEP (Beck et al., 2017) dataset can be obtained from GloH2O: https://www.gloh2o.org/mswep/. Data from the MOSAiC expedition are available at: https://mosaic-expedition.org/mosaic-data/.

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